MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLE FS68

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.

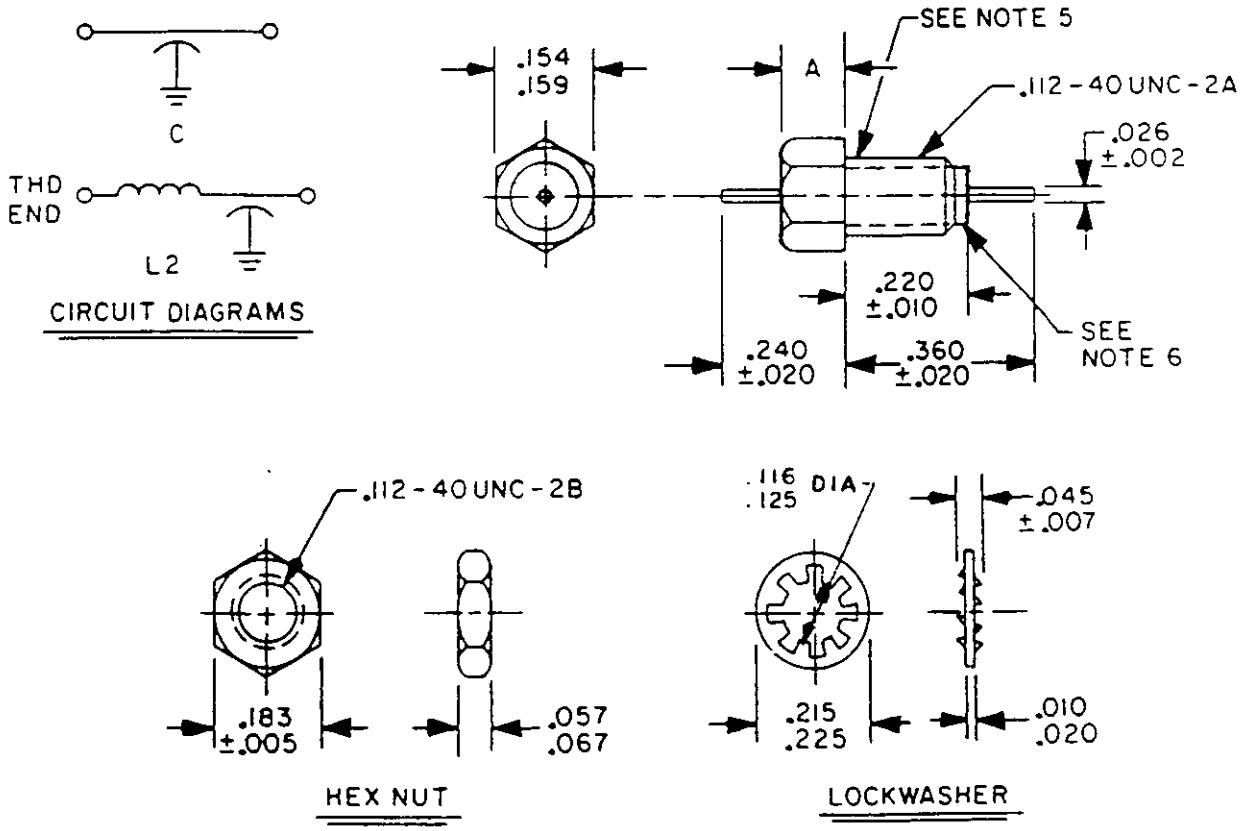


FIGURE 1. Case and hardware dimensions and circuit diagrams.

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Dash number M28861-	A dimension
003	.145 ±.030
001,002,004 through 008	.120 ±.005

Inches	mm	Inches	mm
.002	0.05	.116	2.95
.005	0.13	.120	3.05
.007	0.18	.125	3.18
.010	0.25	.145	3.68
.020	0.51	.154	3.91
.026	0.66	.159	4.04
.030	0.76	.183	4.65
.045	1.14	.215	5.46
.057	1.45	.220	5.59
.067	1.70	.225	5.72
.112	2.84	.240	6.10
		.360	9.14

NOTES:

1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. One and one-half imperfect threads allowed .030 inch (0.76 mm) maximum.
6. One imperfect thread allowed, .030 inch (0.76 mm) maximum.
7. Recommended mounting torque: 32 ounce-inches ±4 ounce-inches.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1 gram maximum.

Case: Steel, UNS 10100 or 10180 in accordance with ASTM A108.

Case finish:

Class B filters, S only (silver plated), in accordance with MIL-F-28861.

Class S filters, S or G only (silver plated or gold plated), in accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 5 amperes maximum.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3.0 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm-microfarads or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarads or 10,000 megohms minimum, whichever is less.

Insertion loss:

At +25°C: In accordance with table I.

At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.1 volt maximum.

DC resistance: 0.02 ohm maximum.

Seal: In accordance with MIL-F-28661. Leakage rate for class S filters shall not exceed 1×10^{-7} atm cm^3/s.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Quality assurance provisions: In accordance with MIL-F-28861.

Part or Identifying Number (PIN): The PIN shall be as follows:

M28861/18-XXX-X-X

XXX	X	X
Dash no. per figure 1 and table I	Case finish	Product assurance level per MIL-F-28861 and table I

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TABLE I. Electrical characteristics.

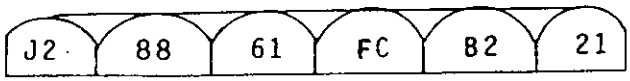
Dash number	Circuit	Product assurance level		Rated voltage volts dc	Capacitance (μF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
		Class				1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
		B	S								
001	C	X	X	100	.027	10	30	39	45	65	70
002	L2	X	X	100	.027	10	30	50	54	70	70
003	L2	X		100	.045	14	37	50	58	70	70
004	C	X	X	200	1,000 pF		4	20	25	40	55
005	C	X	X	200	5,000 pF		15	34	41	50	55
006	C	X		200	.01	4	21	35	42	55	70
007	L2	X		200	.01	4	21	35	38	65	70
008	C	X	X	50	.050	15	33	44	50	67	70

1/ For C circuits, insertion loss measurements shall be made under no load. For L2 circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz; above this frequency range, insertion loss measurements shall be made under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

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Marking: Filters shall be marked, as a minimum, with an abbreviated military PIN as shown on figure 2. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.



J - Abbreviation for JAN
28861 - Abbreviation for M28861
F - Indicates specification sheet 18
C - Code indicating dash number as follows:

Code	Dash number	Code	Dash number
A	001	E	005
B	002	F	006
C	003	G	007
D	004	H	008

B - Code indicating product assurance level (for example, B indicates class B).
221 - Date code. First digit indicates the year (for example, 2 indicates 1992). The last two digits indicate the week.

FIGURE 2. Example of marking for the abbreviated military PIN on the hex flats - expanded view.

Cataloging information: Dash numbers 001, 004 through 006, and 008 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 002, 003, and 007 shall be cataloged under FSC 5915 as radio frequency interference filters.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0103)

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